

REDDING 12 WNW, CA, USA

WMO: 744970

Lat: 40.6507N

Lon: 122.6068W

Elev: 1418

StdP: 13.96

Time Zone: -8.00 (NAP)

Period: 03-19

WBAN: 04222

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.4	31.0	9.3	9.3	54.1	14.2	11.9	51.6	10.9	46.3	8.9	44.8	4.1	N/A	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.4	103.2	65.4	100.0	64.3	96.8	63.7	68.4	95.5	67.2	94.0	65.9	92.2	3.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.7	80.4	72.9	57.6	74.7	73.4	55.8	69.9	73.9	33.3	96.2	32.2	93.6	31.2	92.0	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.9	7.8	6.9	DB	24.4	107.7	3.2	2.8	22.1	109.7	20.3	111.4	18.5	113.0	16.2	115.0	
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	46.0	47.5	51.3	55.8	64.1	73.2	80.1	72.7	61.7	51.3	44.2		
		DBStd	14.22	6.54	6.41	7.47	7.92	7.33	7.94	5.49	4.95	7.22	7.32	7.26	5.67	
		HDD50	644	160	113	74	30	2	0	0	0	5	68	192		
		HDD65	3160	589	489	428	294	106	24	0	21	150	414	644		
		CDD50	4504	36	44	113	202	438	696	933	874	680	368	107	13	
		CDD65	1544	0	0	2	16	77	270	468	409	251	48	3	0	
		CDH74	20894	4	4	80	318	1142	3497	6395	5479	3235	698	42	0	
		CDH80	11701	0	0	6	76	425	1886	3960	3288	1812	243	5	0	
(14)	Wind	WSAvg	3.9	3.8	3.8	3.8	3.8	3.9	4.1	4.0	4.1	4.2	4.1	3.8	3.7	
(15)	Precipitation	PrecAvg	45.8	8.8	7.0	5.2	3.5	2.4	1.0	0.2	0.3	0.2	2.1	5.6	9.8	
(16)		PrecMax	77.2	26.4	21.1	16.6	8.6	7.8	3.0	0.7	1.3	1.0	5.8	12.7	23.0	
(17)		PrecMin	27.1	1.7	0.7	0.4	1.2	0.1	0.0	0.0	0.1	0.0	0.2	1.3	1.7	
(18)		PrecStd	14.0	5.7	5.2	4.2	1.9	2.4	1.0	0.2	0.4	0.3	1.7	3.5	6.6	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	72.4	73.7	80.8	86.3	94.2	104.5	107.3	105.9	102.6	92.8	79.6	67.3		
(20)		MCWB	50.4	53.1	55.4	59.1	61.9	66.3	68.0	65.6	63.2	60.5	55.8	48.6		
(21)		DB	67.0	68.8	76.3	82.0	89.0	99.3	104.0	101.5	98.7	86.3	72.7	61.8		
(22)		MCWB	48.5	51.1	54.4	57.6	60.9	64.0	66.1	64.3	61.7	58.1	52.9	46.9		
(23)		DB	61.4	64.4	71.0	77.7	85.0	95.2	100.6	98.0	94.9	81.8	68.0	56.7		
(24)		MCWB	46.9	49.0	52.4	56.2	59.7	63.1	65.3	64.3	61.2	56.9	51.3	45.2		
(25)		DB	56.3	59.1	65.7	72.1	80.8	90.9	97.1	94.9	91.2	77.2	62.6	52.6		
(26)		MCWB	45.7	46.7	50.2	54.0	58.3	62.3	64.3	63.7	61.2	55.7	49.4	46.0		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	53.3	55.4	57.7	63.0	65.5	69.8	70.7	69.1	67.4	62.6	59.2	53.4		
(28)		MCDB	62.9	69.5	72.6	80.7	87.2	96.0	100.1	97.3	91.6	79.1	72.7	57.5		
(29)		WB	50.6	53.4	55.2	60.1	63.4	67.7	68.8	67.6	65.3	60.6	56.6	51.6		
(30)		MCDB	59.7	61.9	68.2	78.5	83.5	93.9	96.6	94.6	89.9	80.9	68.0	53.8		
(31)		WB	49.3	51.7	53.8	57.9	61.4	65.9	67.4	66.3	63.3	59.1	53.8	49.7		
(32)		MCDB	57.2	60.3	64.8	74.6	80.5	91.6	95.4	93.1	88.8	78.2	63.2	52.5		
(33)		WB	47.7	49.7	52.2	55.7	59.9	64.1	66.1	65.0	61.9	57.2	52.0	47.4		
(34)		MCDB	55.3	57.9	61.5	70.7	78.0	89.3	93.3	91.6	87.0	73.0	60.6	52.9		
(35)	Mean Daily Temperature Range	MDBR	18.0	18.7	19.9	22.1	24.8	27.8	31.4	31.8	30.4	24.8	18.7	15.8		
(36)		5% DB	26.5	27.8	29.1	30.8	32.1	34.2	35.5	35.5	36.3	32.4	25.8	24.1		
(37)		MCWBR	13.3	13.4	13.4	13.4	13.1	11.9	12.1	11.7	12.7	12.6	11.9	13.5		
(38)		5% WB	19.9	21.3	22.9	26.7	29.4	31.7	33.2	31.9	31.7	27.4	20.9	14.9		
(39)		MCWBR	11.2	11.5	11.8	12.6	12.7	11.6	11.7	11.3	11.6	11.9	11.0	9.6		
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.303	0.307	0.319	0.329	0.327	0.334	0.325	0.318	0.313	0.295	0.302		
(41)		taud	2.568	2.562	2.572	2.525	2.531	2.557	2.533	2.549	2.552	2.581	2.602	2.556		
(42)		Ebn at Noon	278	292	302	303	300	300	297	298	294	284	277	266		
(43)		Edh at Noon	22	26	29	32	32	32	32	31	29	25	21	21		
(44)	All-Sky Solar Radiation	RadAvg	632	917	1302	1790	2224	2539	2583	2295	1827	1229	739	528		
(45)		RadStd	120	131	169	183	161	145	82	81	77	114	85	112		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (18 neighbors)	N/A	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	+182	+31

Nomenclature: See separate page